SUPPORTING ADULT GUIDANCE

How fast is the line changing?

Straight Lines and Connected Representations. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner isolates and interprets the **gradient m** — the constant rate of change — while the intercept is held fixed. The emphasis is on reading a gradient in two parts: its **sign** (direction) and its **size** (steepness), including **negative** and **zero** gradients, and on seeing the gradient as the same 'constant step' met earlier in a table.

Driving Question: *What does m control when the intercept stays fixed?*

The mathematics, in plain terms

The **gradient m** is how much y changes each time x increases by 1 — the line's **rate of change**. It is exactly the constant step (the first difference) seen in a table: for **$y = 3x + 1$**, y goes up by 3 every step, so the gradient is 3. A gradient has two parts. Its **sign** gives the **direction**: a **positive** gradient rises as x increases, a **negative** gradient falls, and a gradient of **0** is a flat (horizontal) line. Its **size** gives the **steepness**: a gradient of 5 is steeper than 2, and — this is the point that catches people out — a gradient of –**4 is steeper than +2**, because steepness is about size, not sign. Changing m **pivots** the line about its intercept: the crossing point stays put while the line tilts. On a graph you can read the gradient with a step: go **1 across** and count how far **up** (or down) the line goes — that rise is m. In the real world a gradient is any steady **rate**: speed (distance per hour), a wage (pay per hour), a tank draining (a negative rate). This lesson holds the intercept fixed so the effect of m is clean; the next few lessons use two points to find a gradient and then write the whole equation.

Sign and size

The heart of this lesson is reading a gradient as two things at once: a **sign** (which way the line leans) and a **size** (how steep). Keeping these separate is what prevents the classic error that a big *negative* gradient is somehow 'small'. It is worth grounding the gradient in the '1 across, so-many up' step, and in the constant rate from the earlier table lesson — that keeps 'gradient' from becoming an abstract word and ties it to change the learner can see. Negative gradients lean on directed number, so a quick reconnection there pays off; the falling line is often the first place a learner really feels what a negative rate means. Protect the observation that the intercept does not move as m changes — the line pivots — which mirrors the previous lesson and sets up writing equations soon.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Gradient = the rate: change m and read the table's constant step — the '1 across, m up' rate of change.	~12 min

Movement	Gradient Isolator: intercept locked, drag m and watch the line pivot — rising, falling or flat.	~12 min
Reflection	Read and compare gradients: direction from the sign, steepness from the size.	~10 min
Creativity	Model a real rate: match a real situation's rate to a gradient, including negative rates.	~5 min
Rest	A pause after feeling one number control a line's whole steepness and direction.	~4 min
Nutrition	Contextual (Mode A): speed, wages, price-per-litre, a draining battery — rates are gradients.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking a big negative gradient is 'small' or 'less steep'

Repair: “Steepness is the size of m. Which is steeper — a gradient of –4 or +2? Compare their sizes.”

Reading a falling line as having a positive gradient

Repair: “As x increases, does y go up or down? Down means the gradient is negative.”

Confusing the gradient with the intercept

Repair: “Which number changes the steepness, and which changes where the line crosses? Change m — does the crossing point move?”

Thinking a flat line has no gradient (rather than gradient 0)

Repair: “How much does y change as x increases here? If it never changes, the rate — the gradient — is 0.”

Getting the rise-over-run upside down (run over rise)

Repair: “Go 1 across first, then count the up (or down). The gradient is the change in y for a change of 1 in x.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: reading coordinates, and directed (positive and negative) numbers. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the sign, the size and the step with the learner:

- “As x goes up by 1, how far does y go — and up or down?”
- “Is this gradient positive, negative or zero? So which way does the line lean?”
- “Which of these two lines is steeper? Compare the sizes of their gradients.”
- “Can you show the gradient as a ‘1 across, so-many up’ step on the graph?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Is the gradient the same as the rate of change?”
Yes — they are two names for the same thing: how much y changes for each 1 that x increases. Grounding gradient in a real rate (speed, cost per item) often makes it click.

“Why do negative gradients cause trouble?”
They combine two ideas at once — a direction (falling) and a size (steepness) — and they lean on directed number. A short reconnection on negatives, and plenty of falling-line examples, settle it.

“How does this set up the next lessons?”
Once a learner can read a gradient as a rate and a direction, the next step is finding a gradient from two points, and then combining gradient and intercept to write the whole equation $y = mx + c$.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.

Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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